

Chandan Kumar Singh

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Summary

Software Engineer with 1+ year of experience designing scalable distributed systems using Java, Spring Boot, Kafka, gRPC, and AWS. Proven track record in building microservice backends, API gateways, and event-driven architectures that cut latency by 40-60% and improve throughput. Skilled in securing services, automating workflows, and applying AI/ML techniques to enhance system reliability and performance.

Experience

Tata Consultancy Services

Software Engineer — Java, Spring Boot, Microservices

Jan 2025 – Present

- Designed and owned RESTful APIs for banking workflows, reducing average response latency by 40% and boosting platform scalability by 50% through service-layer optimizations.
- Architected a centralized API Gateway with secret-key authentication and JWT token-reuse, cutting inter-service overhead by 60% and enhancing distributed system reliability.
- Engineered secure integrations with external banking systems using AES-256 encryption and defenses against MITM, replay, and tampering attacks, achieving zero security incidents.
- Automated internal processes, eliminating 64% of manual effort and accelerating team release velocity.
- Built a high-throughput bulk-upload pipeline handling thousands of concurrent records per request, increasing system throughput by 35% with zero data-consistency issues.

Projects

Clothify Shopping Application | Java, Spring Boot, Hibernate, PostgreSQL, React.js, gRPC, Kafka, Docker, Kubernetes, AWS

- Built a scalable e-commerce backend with API Gateway routing, Eureka service discovery, and domain-separated microservices for product, user, and order management.
- Implemented low-latency gRPC + Kafka-based asynchronous communication with JWT security; containerized and deployed on Kubernetes (AWS EC2) via Jenkins CI/CD, supporting 100+ concurrent users and reducing latency by 30%.

QuantPulse AI | Python, Java, Spring Boot, FastAPI, React.js, XGBoost, Random Forest, FinBERT, PostgreSQL, Redis, WebSockets

- Created an ensemble stock-prediction engine combining XGBoost, Random Forest, and Logistic Regression on 60+ technical indicators with FinBERT-based news sentiment, improving accuracy over single-model baselines.
- Architected a real-time streaming stack (Spring Boot + FastAPI + React + WebSocket) with Redis caching and resilient Groww API integration using exponential backoff.

Technical Skills

Languages: Java, Python, C++, SQL, JavaScript, TypeScript

Frameworks: Spring Boot, Spring Security, Hibernate, gRPC, Kafka, REST API, FastAPI, React.js

Databases: PostgreSQL, MySQL, Redis, FAISS, ChromaDB

Cloud: AWS (EC2, S3), Docker, Kubernetes, Jenkins

Tools: Git, GitHub, Jenkins, Docker, Kubernetes, NumPy, Pandas, Scikit-learn, HuggingFace, Transformers, LangChain, SVN

Education

Motihari College of Engineering, Motihari

B.Tech in Computer Science and Engineering (CSE)

Oct 2020 – Aug 2024